



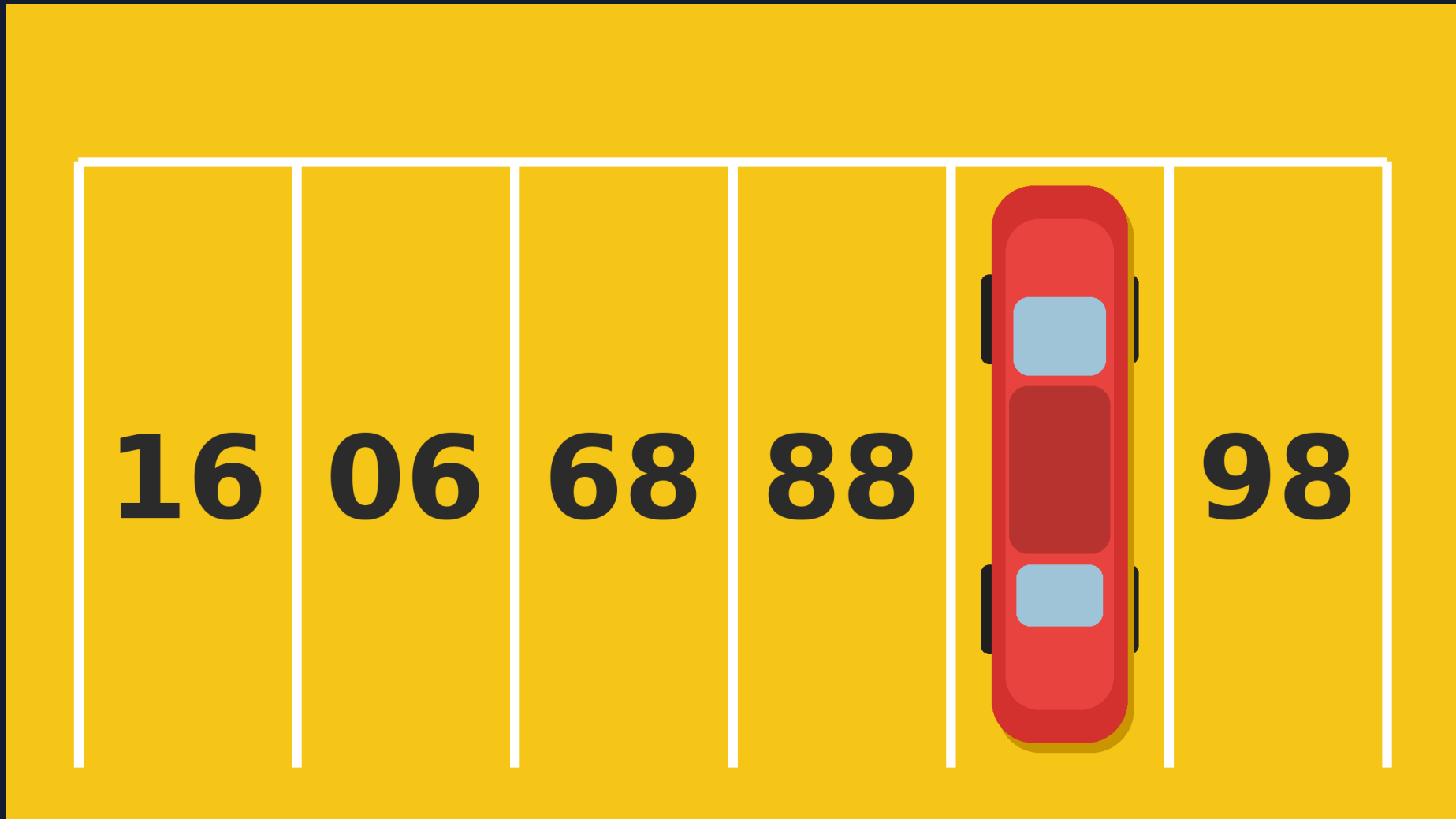
A HANDS-ON WORKSHOP

Critical Thinking & Problem Solving

See clearly. Question deeply. Decide better.

Because the most expensive words in any organisation are: “We assumed...”

What's the number of the parking space?



REVEAL

Turn the problem upside down: it's 87.

86



88

89

90

91

The lesson

Nothing was hidden. The numbers 86–91 were in front of you the whole time — viewed from the driver's side.

Hard problems often stay hard only because we refuse to change the angle we look from.

THE LENS WE FORGET WE'RE WEARING

A White Wall & Your Pink Glasses



**Wear pink glasses long enough,
and every white wall looks pink.**

*Sooner or later, you forget the glasses exist — and swear the wall itself
is pink.*

Your glasses

Past experience, role, culture, incentives, mood — the invisible filter on every fact you see.

The wall

Reality: the data, the customer, the situation as it actually is.

The trap

We debate the colour of the wall. We almost never inspect the glasses.

The skill

Critical thinking = noticing the lens: “What am I bringing to this that isn’t in the data?”

Which way is this bus travelling?

Which way is this bus travelling?



Left or Right?

REVEAL

The clue is what's absent — the doors.

Which way is this bus travelling?



No doors visible → you're seeing its **RIGHT** side.
In India, doors are on the **LEFT** — so it's travelling **LEFT**.

The lesson

No doors visible → you're seeing the bus's right side → in India, it must be travelling left.

Evidence isn't only what's present. Ask: "What's missing that should be here?" — the silent customer, the question nobody raised, the risk no one logged.

Is there a problem?

Before solving anything, agree on what a “problem” even is:

CURRENT STATE

Where we are

THE GAP

= the problem

DESIRED STATE

Where we want to be

No gap → no problem. Unclear gap → unsolvable problem. A problem well-stated is a problem half-solved.

The most expensive words in history: “We assumed...”

~50%

of organisational decisions fail — mostly from **HOW** they were decided, not what was decided. Paul Nutt (Ohio State) tracked 400+ real decisions over two decades: failures were driven by rushed framing, solutions imposed early, and evidence gathered to persuade rather than to test.



Mars Climate Orbiter • 1999

\$327M burned up

One team worked in metric, one in imperial. Nobody verified the assumption that “everyone uses the same units.”



Challenger • 1986

7 lives

Engineers flagged the O-ring risk in cold weather. Schedule pressure and “it flew fine before” overrode the warning — normalisation of deviance.



Boeing 737 MAX • 2018–19

346 lives • ~\$20B+

Design assumed pilots would diagnose and counter MCAS within seconds — an assumption about humans, never stress-tested like the hardware.



Global financial crisis • 2008

Trillions, worldwide

An industry priced risk on the assumption that US housing prices don’t fall nationally. The models were precise; the premise was wrong.

The quiet everyday version: 9 of 10 megaprojects overrun their budgets (Flyvbjerg, Oxford) — not from bad engineering, but from optimistic assumptions nobody was assigned to attack.

So what exactly is critical thinking?

The discipline of **actively questioning** what you see, hear and believe — examining evidence, surfacing assumptions and weighing alternatives *before* judging or acting.



It is NOT...

- Being negative or criticising everything
- Slow, academic over-analysis of every choice
- A talent you're born with (or without)
- Only for “big” strategic decisions



It IS...

- Constructive doubt — questioning to improve, not to attack
- A fast habit: 30 seconds of “what am I assuming?”
- A trainable skill that compounds with practice
- Useful in emails, estimates, hiring, debugging, daily calls

WHY IT'S WORTH THE EFFORT

Benefits of critical thinking at the workplace



THINK BETTER

Improved problem solving

You fix causes, not symptoms

Guided decision-making

Choices rest on evidence, not impulse

Greater adaptability

New situations become puzzles, not threats



WORK BETTER TOGETHER

Enhanced communication

Clear thinking → clear speaking & writing

Stronger teamwork

Ideas are examined, not owned — less ego

Conflict resolution

You argue the issue, not the person



GROW FASTER

Enhanced productivity

Less rework from decisions made twice

Better time management

You question urgency, not just react to it

Self-improvement & confidence

Trusting your reasoning, you speak up

The 1st step — a reset of the mindset



FIXED MINDSET

“I’m just not good at this”

Ability is a fixed quantity

Mistakes = proof of inadequacy

So they get hidden, not studied

Feedback feels like attack

So it gets defended, not used

Sticks to what already works

Excellence stays out of reach



GROWTH MINDSET

“I’m not good at this yet”

Ability grows with effort & method

Mistakes = data

The fastest teacher you’ll ever get

Feedback = free consulting

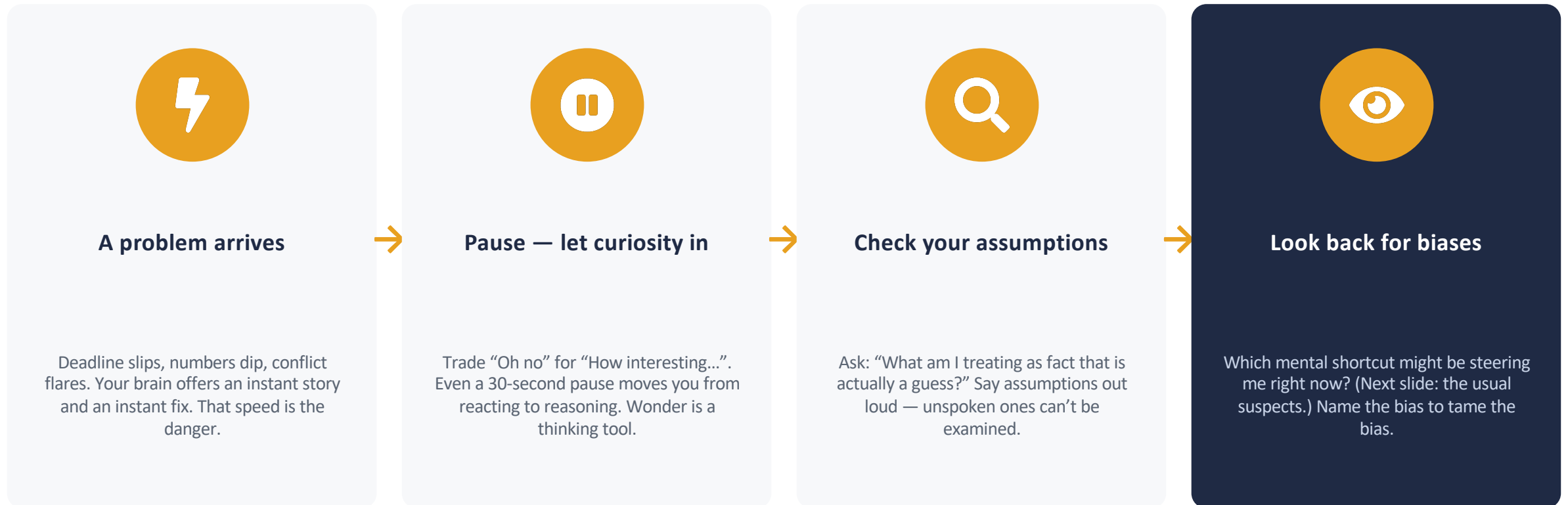
Sought out, weighed, applied

Experiments beyond the comfort zone

Smarter decisions, inviting excellence at every step

You can’t think critically about the world while thinking defensively about yourself.

When a problem arrives: pause before you pounce



Between stimulus and response there is a space. In that space is our power to choose. — attributed to Viktor Frankl

Six biases that bend your thinking daily



Confirmation bias

Hunting for evidence that agrees with you

e.g. Googling “why X is best” after choosing X



Anchoring

Over-weighting the first number or idea heard

e.g. First estimate in a meeting sets the range



Authority bias

Seniority substituting for evidence

e.g. “The MD said so” ends the discussion — Challenger



Sunk cost fallacy

Continuing because of what’s already spent

e.g. “We’ve invested 6 months, we can’t stop now”



Availability bias

Vivid & recent events feel more likely

e.g. One bad vendor story vetoes a whole category



Groupthink

Harmony valued over honest examination

e.g. Silent meetings where everyone “agrees”

You can’t remove biases — you can only build systems that catch them: checklists, devil’s advocates, pre-mortems, “what would change my mind?”

Six steps of critical thinking

1

Identify the issue

State the gap precisely. Vague in, vague out.

2

Gather & evaluate data

Facts before opinions. Check the source, sample & date.

3

Challenge assumptions & biases

What's a guess wearing a fact's clothes?

4

Consider diverse perspectives

Invite the person most likely to disagree.

5

Draw conclusions / decide

Weigh options against explicit criteria — then commit.

6

Act & learn

Execute, measure, and feed what you learn back to step 1.

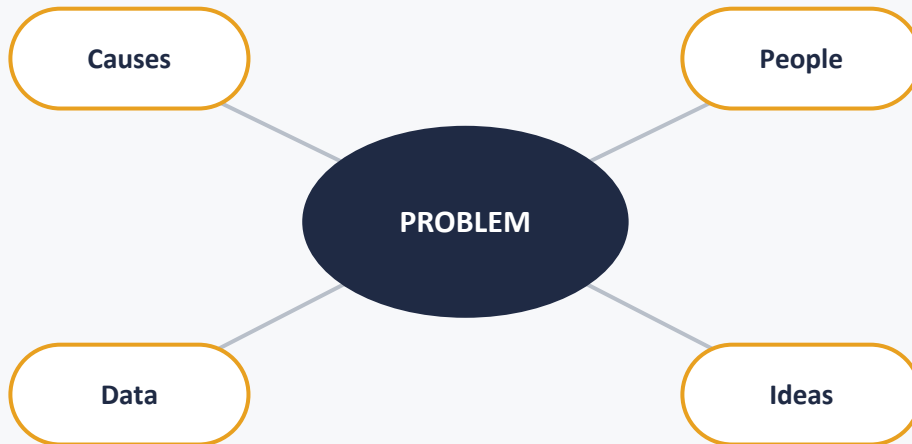
Step 3 is highlighted because it's the step teams skip most — and the one this whole workshop exists for.

Mind Mapping & the Socratic Method



MIND MAPPING

A visual approach: put the problem at the centre, radiate every factor, cause and idea as branches. Seeing it all on one page reveals connections a list hides.



THE SOCRATIC METHOD

Deepen thinking by disciplined questioning — six angles of attack:

Clarify *“What exactly do we mean by...?”*

Probe assumptions *“What are we taking for granted?”*

Demand evidence *“How do we know this is true?”*

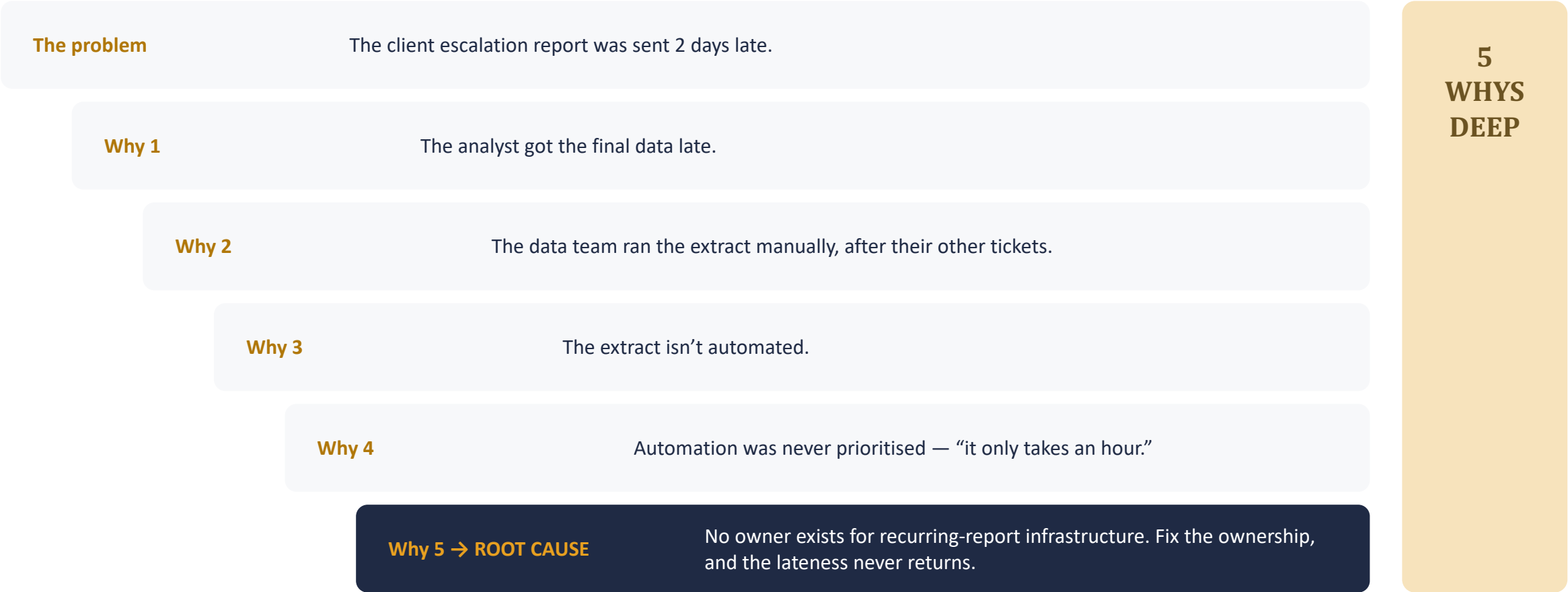
Shift perspective *“How would a customer / rival see it?”*

Test implications *“If true, what follows? Then what?”*

Question the question *“Is this even the right question?”*

Root cause analysis — the 5 Whys

Keep asking “why?” until you hit a cause you can actually fix — usually a process, rarely a person.



Tip: “because someone was careless” is never a root cause — keep going until you reach a system you can change.

SWOT — structure a decision in ten minutes

STRENGTHS

Internal · Helpful

What do we do well? What unique assets, skills or advantages do we hold?

Ask: “What would competitors envy?”

WEAKNESSES

Internal · Harmful

Where do we struggle? What’s missing — skills, resources, processes?

Ask: “What do we avoid talking about?”

OPPORTUNITIES

External · Helpful

What trends, gaps or changes outside could we ride?

Ask: “What’s changing in our favour?”

THREATS

External · Harmful

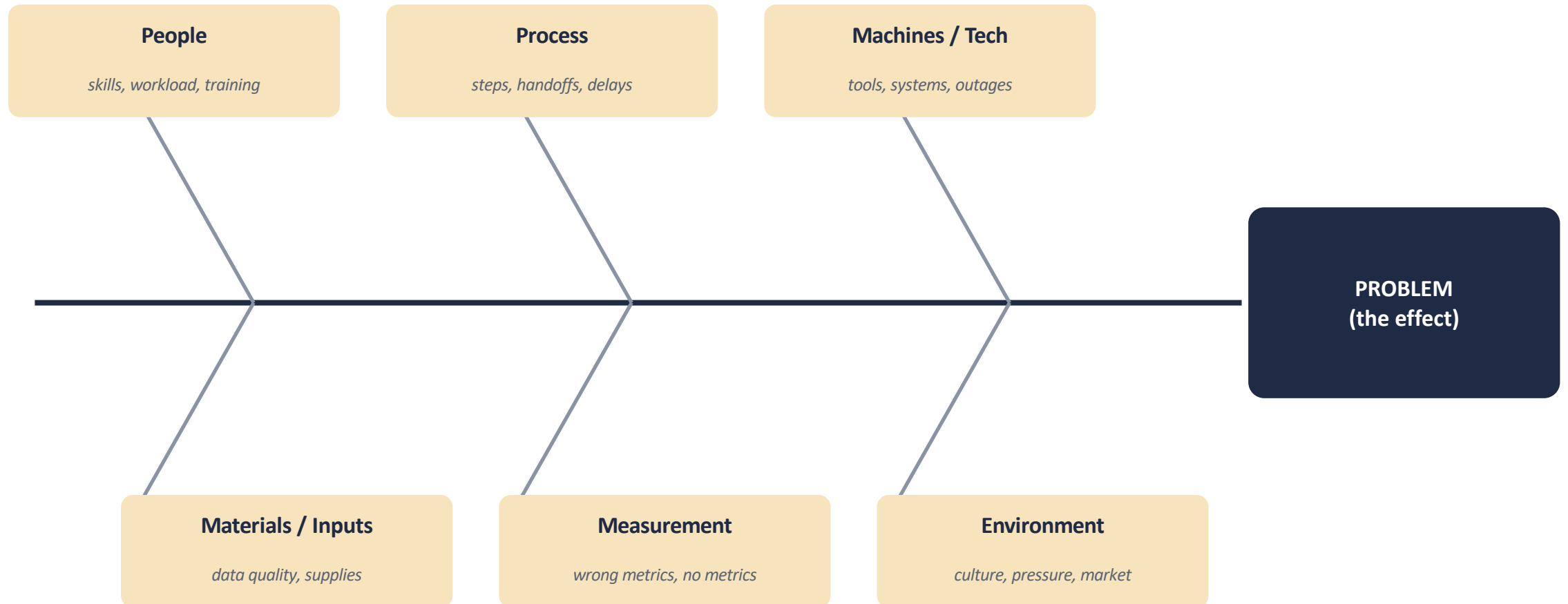
What outside forces could hurt us — rivals, regulation, technology shifts?

Ask: “What keeps us up at night?”

Power move: don’t stop at listing — pair them. Strength × Opportunity = strategy. Weakness × Threat = your biggest risk.

Fishbone / Ishikawa — map every possible cause

When a problem has many suspects, 5 Whys alone is too narrow. The fishbone organises brainstormed causes into categories so nothing hides.



Run it: state the effect → brainstorm causes per bone → circle the 2–3 most likely → verify with data → 5-Whys the winner.

Affinity Diagram & MoSCoW — order from chaos



AFFINITY DIAGRAM

When you're drowning in inputs — feedback, ideas, issues — group them by natural theme and the signal appears.

- 1 Write every item on its own sticky note
- 2 Silently cluster similar notes together
- 3 Name each cluster only AFTER grouping
- 4 Prioritise clusters, not individual items

Silent grouping first — it stops the loudest voice deciding the categories.



MoSCoW PRIORITISATION

M

MUST have

Non-negotiable — without it, the solution fails

S

SHOULD have

Important, but the solution survives without it

C

COULD have

Nice-to-have if time and budget allow

W

WON'T have (now)

Explicitly parked — written down, not forgotten

Which tool, when?



“I’m stuck — I keep seeing it one way”

→ Change the frame: mind map it, invite an outsider, ask the Socratic questions



“Something broke and keeps recurring”

→ Fishbone for breadth of causes → 5 Whys for depth on the strongest one



“Too many issues at once — it’s a mess”

→ KT Situation Appraisal to separate concerns → Affinity to cluster them



“We must choose between options”

→ SWOT each option → KT Decision Analysis with MUST/WANT criteria



“Too much to do, not enough time”

→ MoSCoW — and defend the Won’t-have list in writing



“The plan is set — what could sink it?”

→ KT Potential Problem Analysis / pre-mortem before launch, not after

Tools don’t compete — they chain. Real problem-solving is choosing the right sequence.